

Speakardo Software Requirements Specification (SRS)

Version: 1.0

Product: Speakardo

Category: AI Life Assistant

Status: Living Technical Document

1. Introduction

1.1 Purpose

This document defines the technical and functional requirements for Speakardo, an AI-powered life assistant that enables users to create reminders, manage personal memory, receive notifications, and eventually automate parts of their daily life through natural conversation.

This SRS serves as the source of truth for engineering, design, AI systems, infrastructure, and future product expansion.

2. Product Scope

Speakardo is not a traditional reminder application.

Speakardo is an AI Life Assistant that:

- Understands natural language
 - Creates reminders automatically
 - Learns user behavior
 - Maintains personal memory
 - Delivers reminders through multiple channels
 - Supports collaborative reminders
 - Evolves toward autonomous assistance
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3. User Roles

3.1 Guest

Capabilities:

- Landing page access
- Signup
- Login

Restrictions:

- Cannot create reminders
 - Cannot access assistant
-

3.2 User

Capabilities:

- Chat with AI
 - Create reminders
 - Manage reminders
 - Voice interaction
 - Receive notifications
 - Manage profile
 - Invite others
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3.3 Team Member

Capabilities:

- Receive shared reminders
 - Collaborate on reminders
 - Participate in shared workspaces
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3.4 Family Member

Capabilities:

- Receive family reminders
- Share household reminders

3.5 Admin

Capabilities:

- User management
 - System monitoring
 - Abuse prevention
 - Analytics access
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4. Functional Requirements

Module 1: Authentication

Features

- Email signup
- Email login
- Password reset
- Social login
- Google login
- Apple login
- Session management
- JWT authentication
- Refresh tokens

Acceptance Criteria

User can:

- Register account
 - Login securely
 - Logout securely
 - Recover account
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Module 2: AI Chat Assistant

Features

Natural Language Reminder Creation

Examples:

"Remind me to call Ali tomorrow at 5"

"Wake me at 7 AM"

"Pay electricity bill every month"

AI Responsibilities

Extract:

- Task
- Date
- Time
- Recurrence
- Priority

Generate confirmation:

"Got it. I'll remind you tomorrow at 5 PM."

AI Parsing Pipeline

Level 1

Rule-based extraction

Tools:

- Regex
 - DateParser
 - Duckling
-

Level 2

AI fallback

Providers:

- OpenAI
- Gemini
- Claude

Used when confidence is low.

Module 3: Reminder Management

Reminder Types

One-Time Reminder

Example:

Call Mom at 5 PM

Recurring Reminder

Examples:

Every day

Every week

Every month

Custom schedules

Location Reminder

Examples:

Remind me when I reach office

Remind me when I leave home

Context Reminder

Examples:

Remind me after my meeting

Remind me before my flight

Smart Reminder

AI determines optimal time.

Reminder Actions

Create

Update

Delete

Archive

Complete

Snooze

Reschedule

Duplicate

Share

Module 4: Notifications

Notification Channels

Push Notification

Android

iOS

Web

Voice Notification

TTS reminder playback

Email Notification

Optional

SMS Notification

Premium feature

AI Call Reminder

Premium feature

Example:

"Hello Nouman.

This is your reminder.

Take your medicine."

Features:

- Human-like AI voice
 - Personalized voice
 - Multiple languages
 - Retry if unanswered
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Module 5: Shared Reminders

Shared Reminder System

Users can assign reminders to others.

Example:

Manager → Employee

Parent → Child

Husband → Wife

Friend → Friend

Shared Reminder States

Pending

Accepted

Declined

Completed

Expired

Features

Assign reminder

Transfer reminder

Track completion

Send follow-ups

View history

Module 6: Team Workspaces

Workspace Features

Create team

Invite members

Assign reminders

Shared schedules

Shared tasks

Shared notifications

Team analytics

Roles

Owner

Admin

Member

Viewer

Module 7: Voice Assistant

Features

Voice Input

Speech-to-text

Voice Commands

Hands-free usage

Wake words

Future:

"Hey Speakardo"

Supported Languages

English

Urdu

Arabic

Hindi

Future multilingual support

Module 8: AI Memory System

Personal Memory

Store:

Important dates

Preferences

Relationships

Health information

Personal notes

Life events

Examples

"My mother's birthday is June 10."

"I prefer meetings after 10 AM."

Memory Retrieval

"What did I tell you about my mother?"

"When is my next dentist appointment?"

Module 9: Calendar Integration

Integrations:

Google Calendar

Apple Calendar

Outlook

Features

Sync events

Conflict detection

Smart scheduling

Availability checking

Module 10: AI Scheduling Assistant

Examples:

"Find time next week for a dentist appointment."

"Move all meetings after 3 PM."

Capabilities:

Schedule optimization

Conflict resolution

Time blocking

Priority management

Module 11: Autonomous Assistant

Future System

Capabilities:

Book appointments

Send messages

Reschedule meetings

Order services

Manage daily routines

Take actions with permission

Module 12: Analytics

User Analytics

Completed reminders

Missed reminders

Reminder categories

Behavior trends

Productivity metrics

5. Database Design

Users

Fields:

id

email

password_hash

name

avatar_url

timezone

language

created_at

updated_at

Devices

Fields:

id

user_id

device_token

platform

last_seen

Reminders

Fields:

id

user_id

title

description

datetime_utc

repeat_rule

priority

status

source

created_at

updated_at

ReminderAssignments

Fields:

id

reminder_id

sender_id

receiver_id

status

assigned_at

completed_at

Notifications

Fields:

id

user_id

reminder_id

channel

status

sent_at

opened_at

Memories

Fields:

id

user_id

memory_type

content

importance_score

created_at

Teams

Fields:

id

name

owner_id

created_at

TeamMembers

Fields:

id

team_id

user_id

role

AI Interactions

Fields:

id

user_id

prompt

response

provider

tokens

cost

created_at

6. API Requirements

Authentication

POST /auth/register

POST /auth/login

POST /auth/logout

POST /auth/refresh

Chat

POST /chat

POST /voice

GET /chat/history

Reminders

POST /reminders

GET /reminders

GET /reminders/{id}

PUT /reminders/{id}

DELETE /reminders/{id}

Shared Reminders

POST /shared

GET /shared

PUT /shared/{id}

Teams

POST /teams

GET /teams

POST /teams/invite

Memory

POST /memory

GET /memory

DELETE /memory/{id}

Devices

POST /devices/register

DELETE /devices/{id}

7. Non-Functional Requirements

Performance

Reminder creation:

< 2 seconds

AI response:

< 5 seconds

Notification latency:

< 30 seconds

Scalability

Support:

100,000+ users

1M reminders/day

Horizontal scaling support

Security

JWT authentication

HTTPS only

Encrypted passwords

Rate limiting

Secret management

Data isolation

Audit logs

Reliability

99.9% uptime

Retry mechanisms

Dead-letter queue support

Backup system

8. UI/UX Requirements

Design Principles

Futuristic

Premium

Immersive

AI-native

Memorable

Requirements

Unique layouts

3D elements

Animated assistant

Glassmorphism

Dark mode

Micro interactions

Voice-first experience

Responsive design

Accessibility support

9. Technical Architecture

Frontend

Flutter

Backend

Python

FastAPI

Database

PostgreSQL

Cache

Redis

Queue

Celery

AI Providers

OpenAI

Gemini

Claude

Notifications

Firebase Cloud Messaging

Voice

Whisper

ElevenLabs

OpenAI TTS

Infrastructure

Docker

AWS

Cloud Run

Render

Railway

10. MVP Scope

Included:

Authentication

Chat interface

Reminder extraction

Reminder management

Push notifications

Voice input

Basic AI parsing

Device registration

PostgreSQL

FCM integration

Excluded:

AI memory

Team workspaces

AI calls

Calendar integrations

Autonomous actions

Advanced analytics

11. Future Roadmap

Phase 1

AI Reminder Assistant

Phase 2

Voice Assistant

Phase 3

Shared Reminders

Phase 4

AI Memory

Phase 5

Calendar Assistant

Phase 6

AI Phone Calls

Phase 7

Team Collaboration

Phase 8

Autonomous Life Assistant

Phase 9

AI Life Operating System

12. Risks & Assumptions

Risks

High AI costs

Notification delivery failures

User retention challenges

Platform restrictions

Voice infrastructure costs

Privacy concerns

Assumptions

Users prefer natural language over forms

Users trust AI for reminders

Push notifications remain effective

AI accuracy remains high

Mobile-first experience is preferred

Future automation demand will increase

North Star Requirement

The user should be able to say:

"Speakardo, handle it."

And trust that it will.